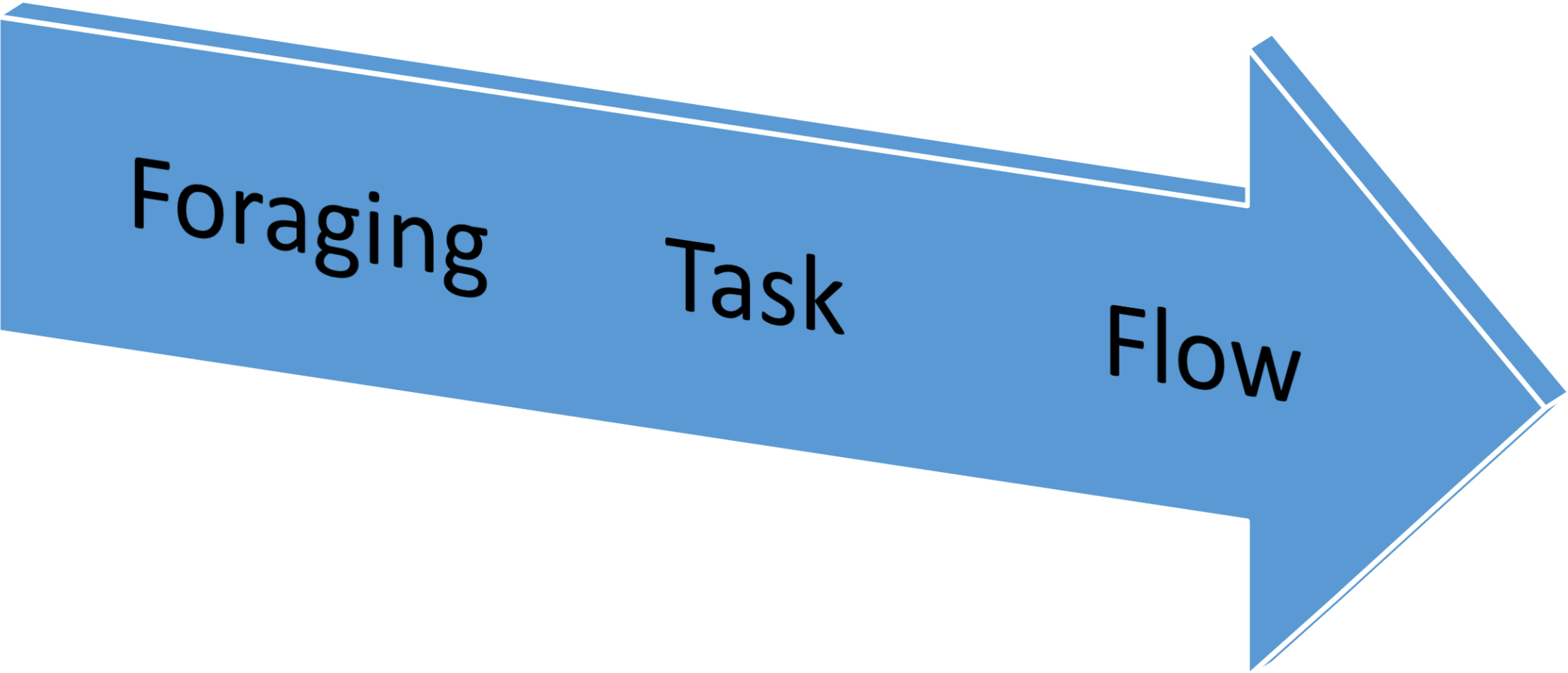


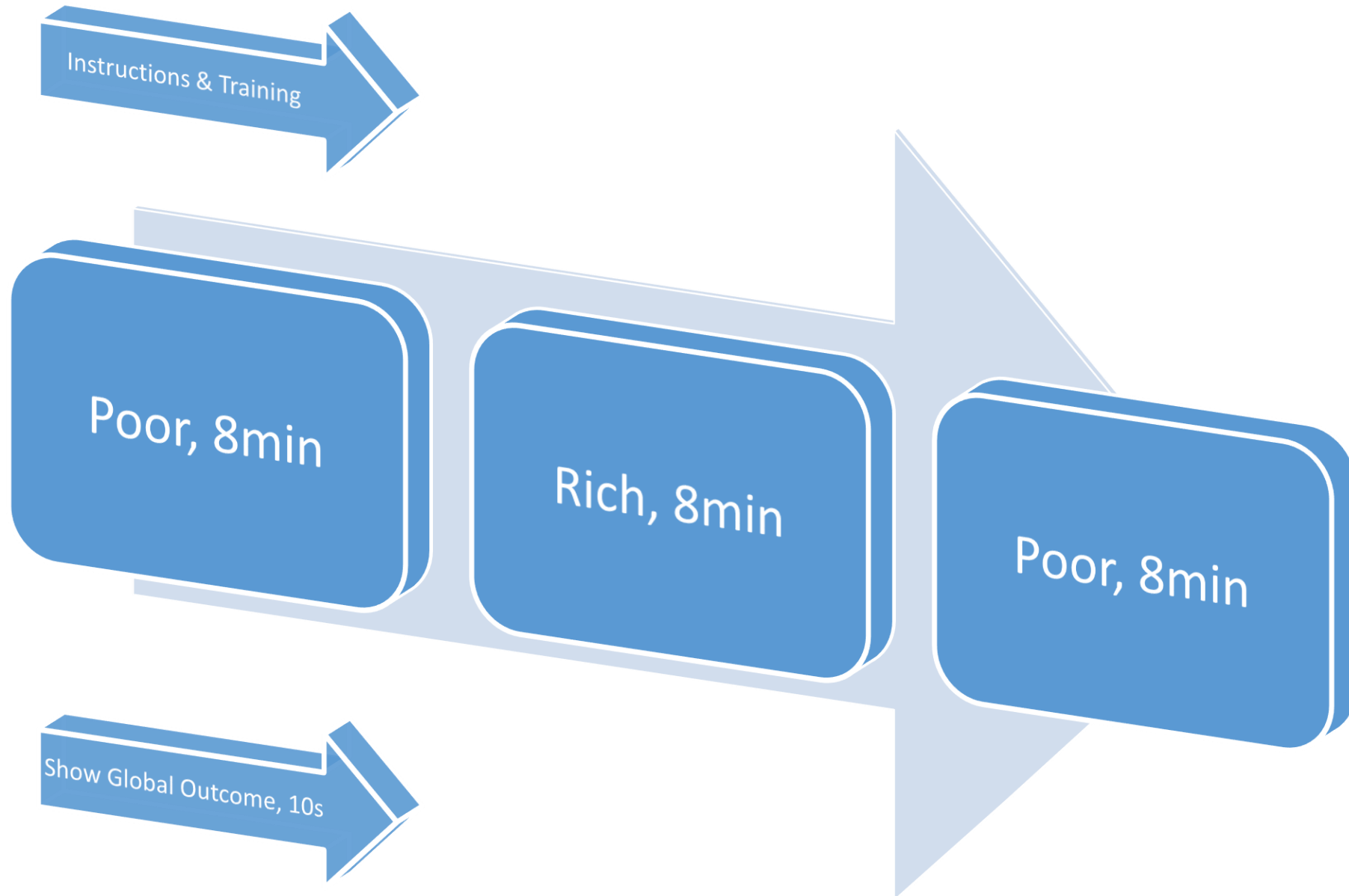


Foraging

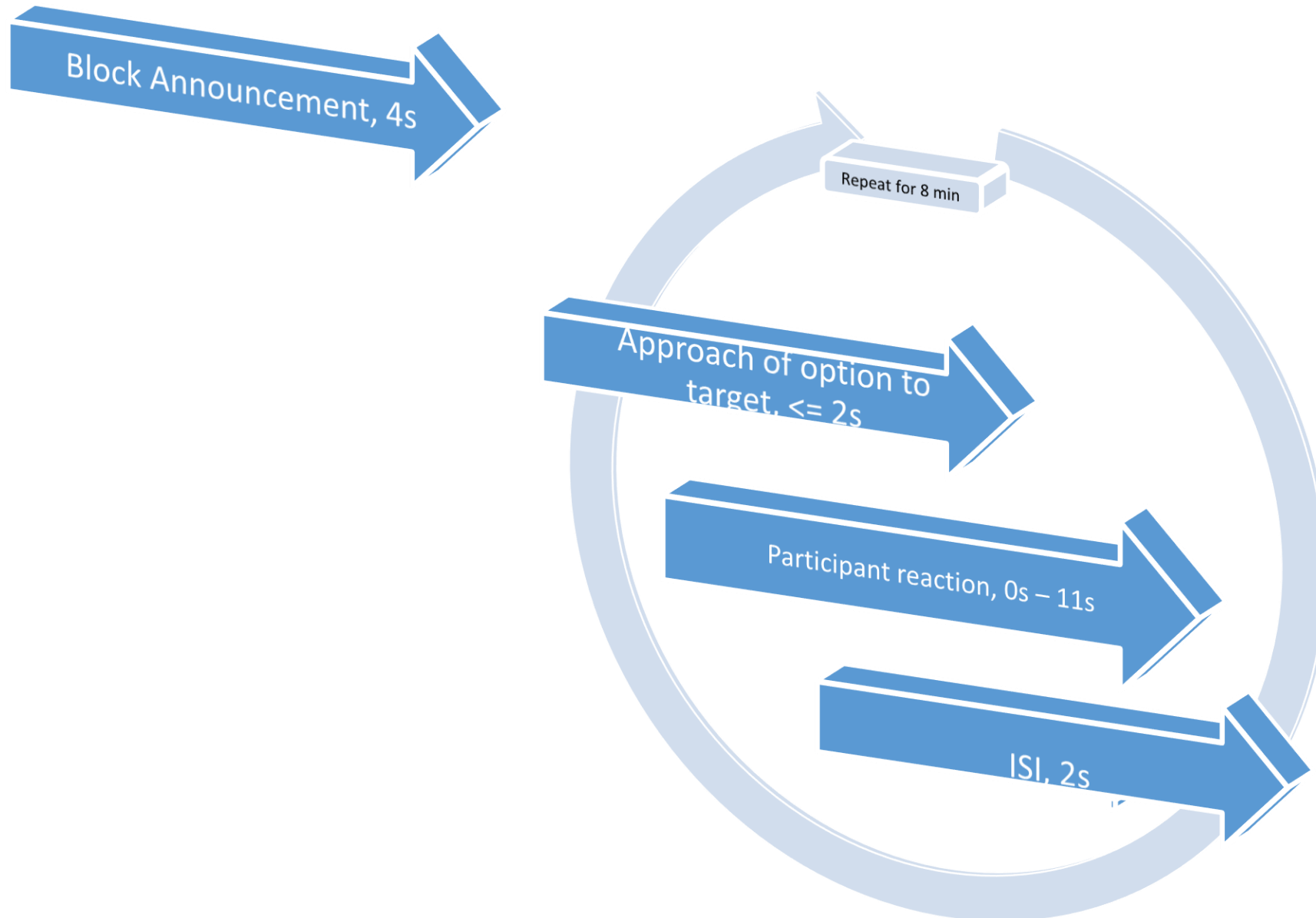
Task

Flow

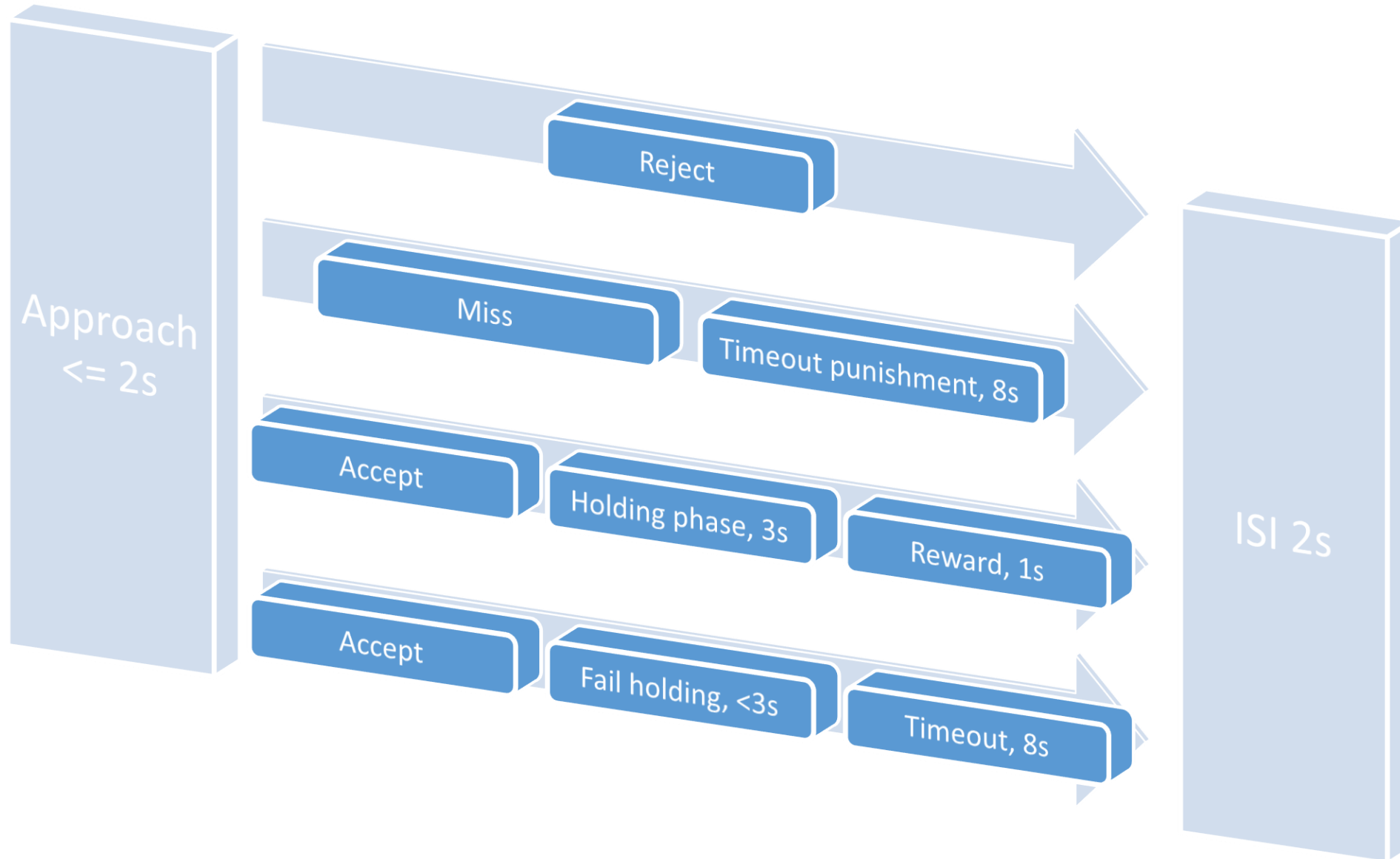
- 1 Session Consists of 3 blocks, ~24 min in total



- Per block



- 4 possible Participant reactions



- Instructions & Training

- Only 2 options, 1 environment, not used in main task

When choosing long training (outside of scanner) you get:

- Full instruction text
- Training runs with short instructions and stepwise increasing difficulty
 1. “catch all aliens”, 60s
 2. “avoid all aliens”, 30s
 3. “maximize fuel as in real game”, 4*60s
- Prospect text for MRI run

When choosing short training (inside of scanner) you get:

- Shortened instruction text
- Training runs with shorter instructions and stepwise increasing difficulty
 1. “catch all aliens”, 30s
 2. “avoid all aliens”, 20s
 3. “maximize fuel as in real game”, 2*60s

- Blocks

Each kind of block has its own properties

- Different background color (also new colors & option-images for 2. session)
- Constitutes a different environment
 - **Rich:** 4 out of 7 options are good (high reward, low effort) + 1 bad option(low reward, high effort) + 2 intermediate (high reward, high effort) & (low reward, low effort)
 - **Poor:** 4 out of 7 options are bad (low reward, high effort) + 1 good option(high reward, low effort) + 2 intermediate (high reward, high effort) & (low reward, low effort)
- Environments, colors and options also differ for training runs, for a fresh learning in main task.

- Per Approach

- A randomly chosen option (represented by an image) approaches one of 2 targets
- Options have different rewards & difficulties
- random choice of option regarding to richness of environment (poor, rich)

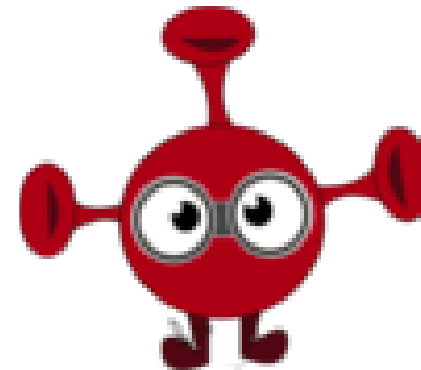
Option 1



Option 2



Option 3

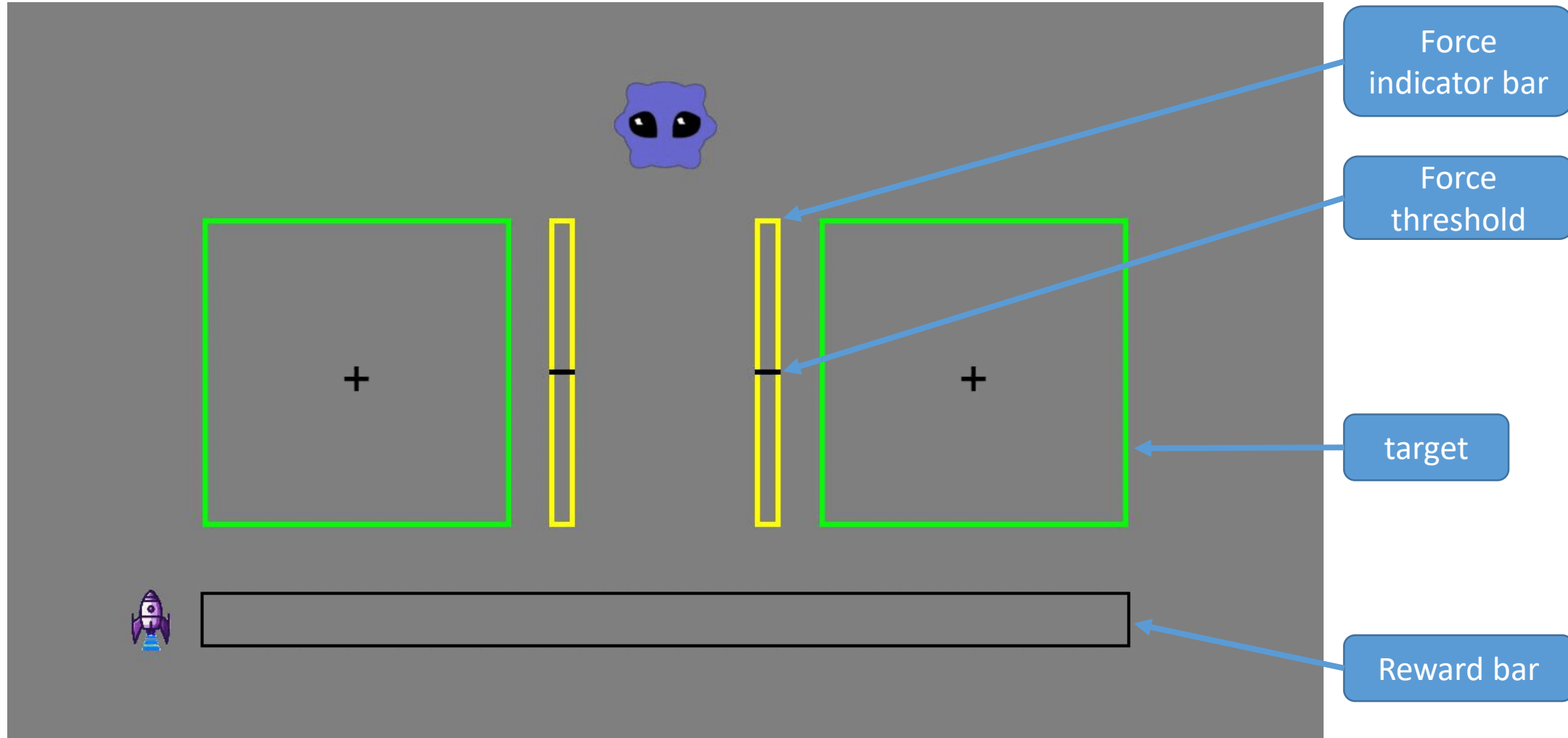


Option 4



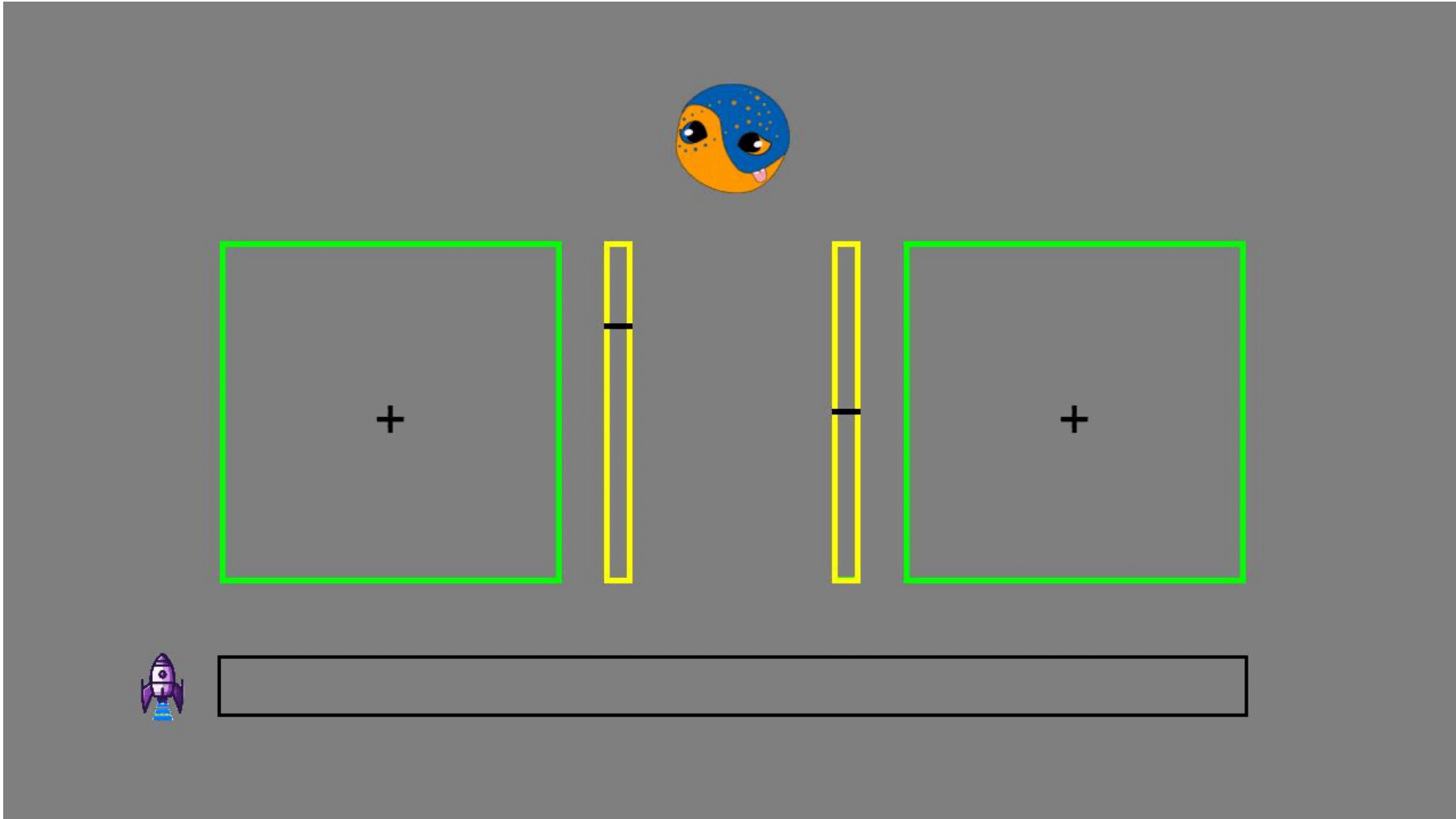
- ## Approach

- For 2 seconds (this is the response window), one option moves towards a randomly chosen target. A participant response ends this phase.



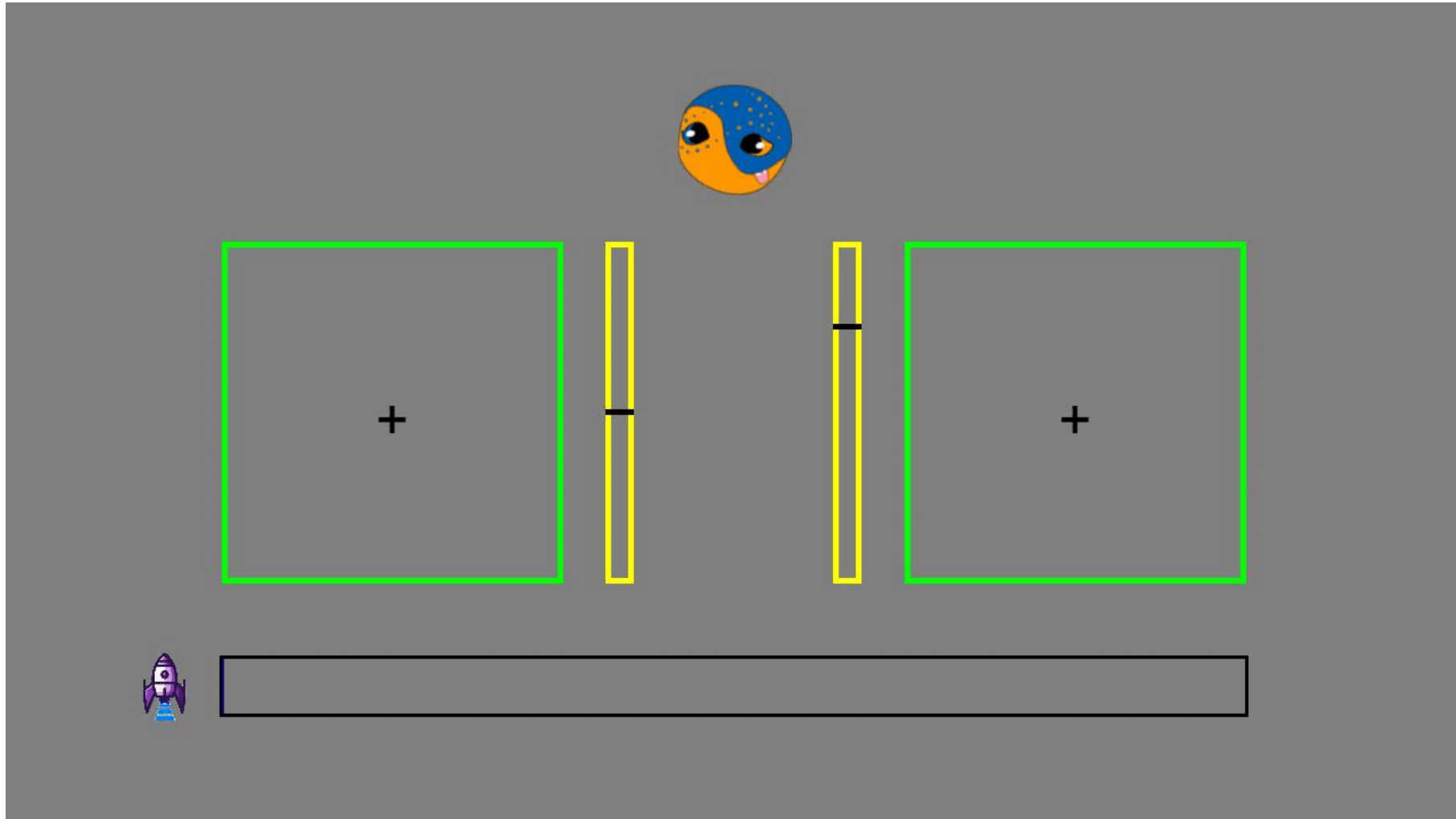
- Participant Reaction - Miss

- Participant misses the response window -> timeout, option stays for 8s in target

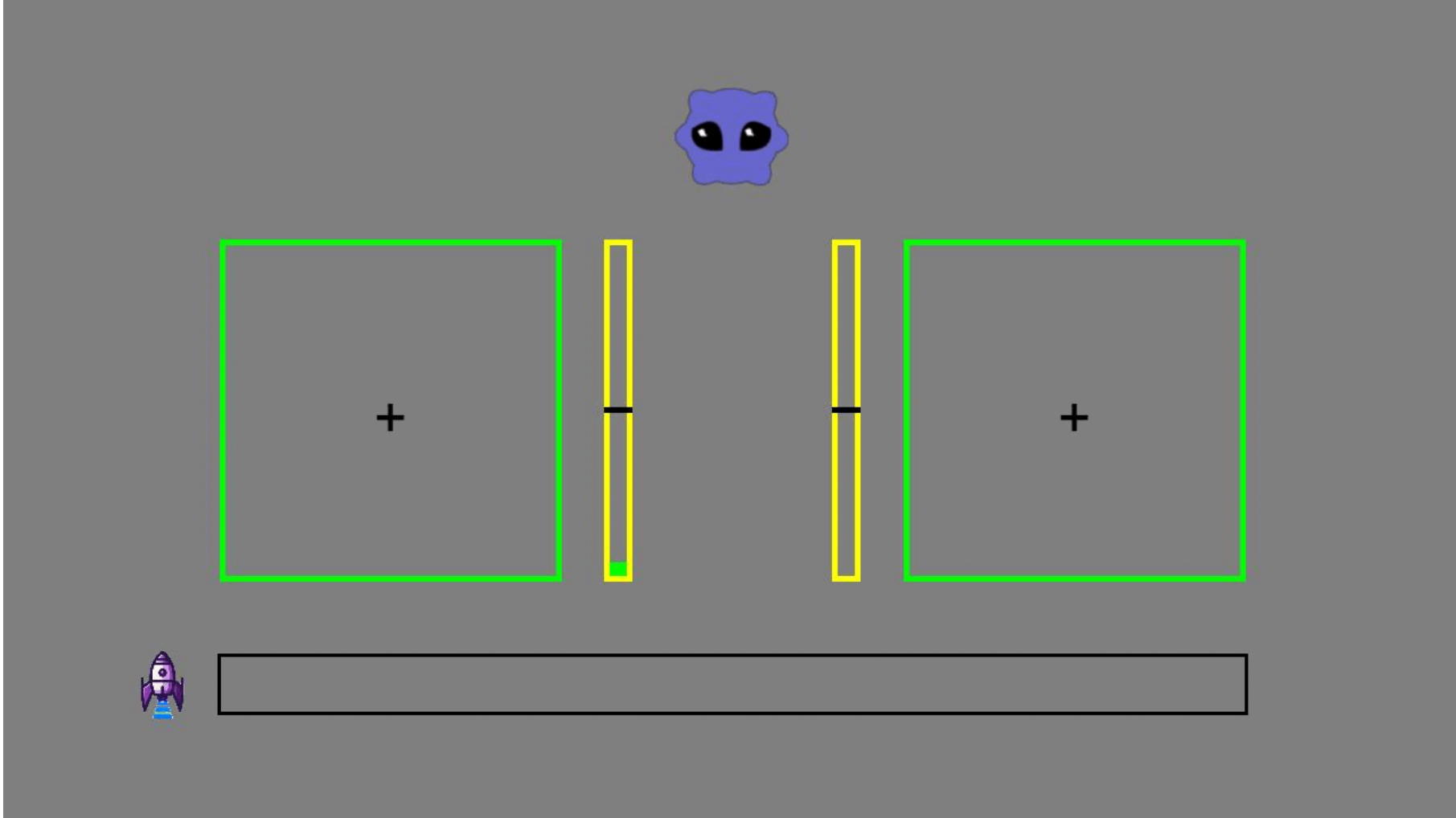


- Participant Reaction - Reject

- Participant chooses the opposite side → skip for now & continue with ISI and next approach



- Participant Reaction – Accept, success
 - Participant accepts option & holds long enough → reward



- Participant Reaction – Accept, fail
 - Participant accepts option & fails to hold long enough → timeout, option stays for 8s in target

